

Actuarial Risk Memo

Claim Frequency & Severity Analysis — Private Passenger Auto Portfolio

Purpose

This memo summarizes a risk analysis of a private passenger auto insurance portfolio, examining how claim frequency and severity vary across key rating variables. The goal is to identify which policyholder and vehicle characteristics are associated with elevated loss potential, in the style of a preliminary ratemaking review.

Data

The analysis uses a public dataset of 67,856 one-year auto insurance policies written in 2004–2005 (the “dataCar” dataset, widely used in actuarial coursework and originating from De Jong & Heller, Generalized Linear Models for Insurance Data). The data includes policy exposure, claim counts, claim costs, and rating characteristics: driver age category, vehicle age, vehicle body type, vehicle value, and geographic area.

Methodology

Consistent with standard actuarial practice, loss potential was decomposed into two components:

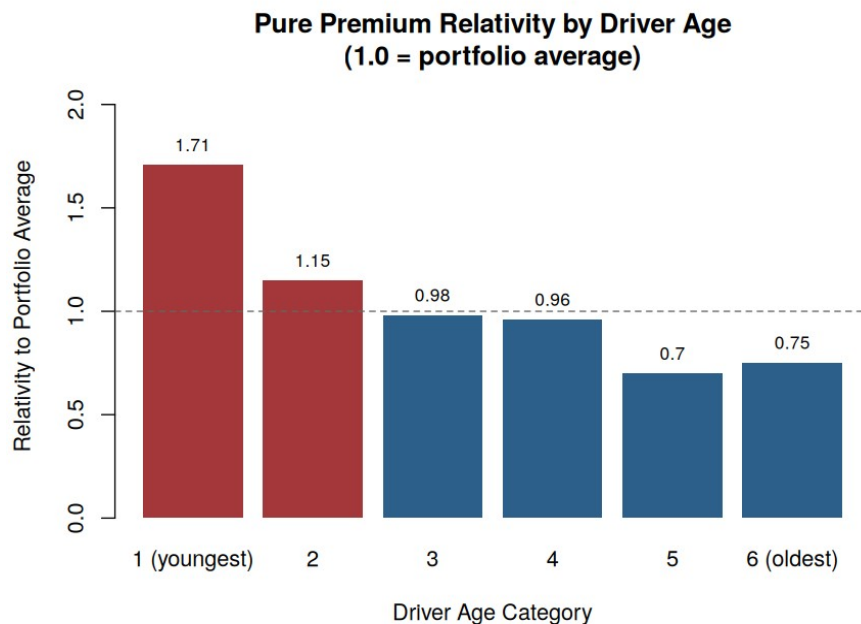
- Claim frequency — claims incurred per unit of exposure (policy-year)
- Claim severity — average cost per claim, conditional on a claim occurring

Frequency and severity were each summarized across three rating variables (driver age category, vehicle age, and geographic area), and combined into a pure premium (expected loss per exposure-year) to assess overall risk relativities relative to the portfolio average.

Portfolio Overview

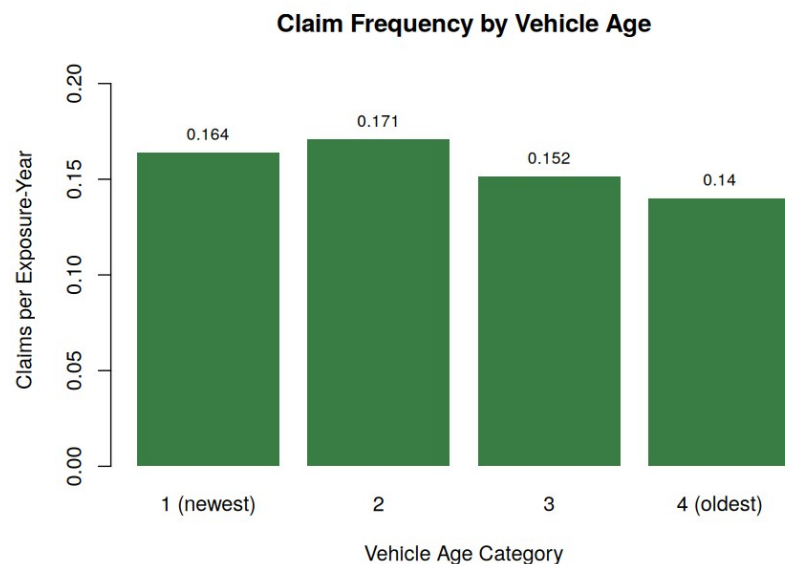
Metric	Value
Policies	67,856
Total exposure	31,801 policy-years
Overall claim frequency	0.155 claims / exposure-year
Average severity	\$1,887 per claim
Portfolio pure premium	\$292.90 per exposure-year

Key Finding 1: Driver Age Is the Dominant Risk Signal



Pure premium relativities by driver age show a clear and steep gradient. The youngest driver category carries a pure premium 71% above the portfolio average, and the second-youngest category runs 15% above average. From the middle age categories onward, relativities fall below 1.0, bottoming out around the second-oldest group at roughly 30% below average. This pattern holds up whether the finding is decomposed into frequency (youngest drivers file claims at nearly double the rate of the oldest categories) or severity (youngest drivers' claims also run noticeably more expensive on average) — both components point the same direction, which is what makes age such a reliable, low-noise rating variable in practice.

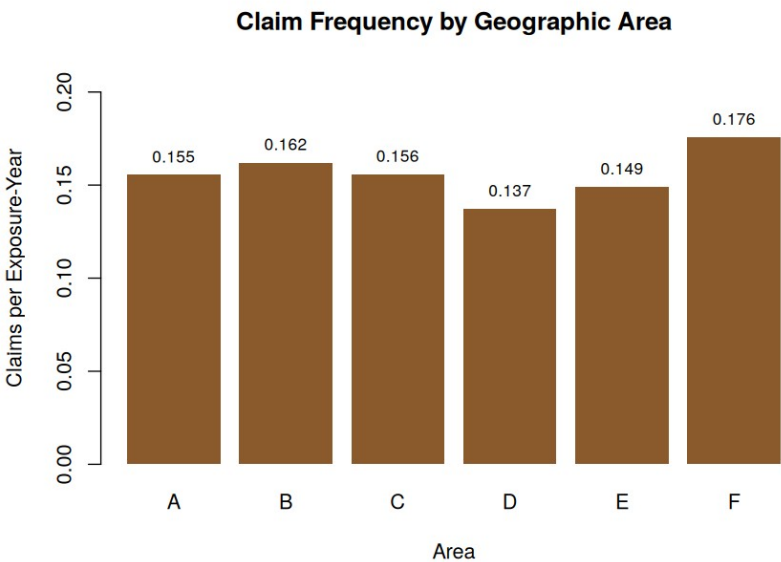
Key Finding 2: Vehicle Age Cuts Against Intuition



Claim frequency actually decreases modestly as vehicles get older (from about 0.16–0.17 for newer vehicles down to 0.14 for the oldest category), while average severity trends upward with vehicle age. A plausible read: newer vehicles may be driven more often or by less experienced household members, while older vehicles that do get into accidents may lack modern

safety features that limit damage costs, or may be costlier to repair due to parts availability. This is exactly the kind of counter-intuitive, non-monotonic relationship that argues for using data over assumption when setting rating factors.

Key Finding 3: Geography Shows a Meaningful but Thinner Signal



Claim frequency by area ranges from about 0.137 to 0.177 across the six geographic zones — a real but more modest spread than driver age produces. Area F stands out with both the highest frequency and, per the severity summary table, the highest average severity, but it also carries the smallest exposure base of any zone, which is a caution flag: with less data behind it, that relativity deserves more scrutiny (or a credibility adjustment) before being trusted at full weight in a rate plan.

Limitations & Considerations

- Single-variable (univariate) relativities shown here can overlap — e.g., younger drivers may also tend toward certain vehicle types — so a full ratemaking exercise would use a multivariate GLM to isolate each factor's independent effect.
- The area-level severity signal is based on a thinner slice of exposure and would benefit from a credibility-weighted estimate rather than being taken at face value.
- Data reflects a single historical period (2004–05) and one market; relativities would need refreshing against current experience before any real-world pricing use.
- Any operational use of driver-age-based rating must also account for applicable regulatory constraints on rating variables, which vary by jurisdiction.

Conclusion

Driver age is the clearest and most actionable risk signal in this portfolio, with both frequency and severity reinforcing a consistent, steep relativity gradient. Vehicle age adds a secondary, non-intuitive signal worth investigating further, and geographic area offers a real but thinner differentiator that would benefit from credibility weighting before being fully trusted. Together, these findings illustrate the core actuarial workflow of decomposing loss cost into frequency and severity to identify and validate rating factors.